Credit risk portfolio

Solution overview | Synthetic bank data | Local handover

14,685 loan accounts	**16.10%** observed default rate	**$601.12m** original exposure

How a risk analyst uses it

Portfolio: filter loan type, region, income and employment; compare exposure, default rates, age, income/employment cuts and repayment by loan age. **Account:** inspect a score, reference-value sensitivities and repayment history. **Anomaly:** review reasons and record a decision. **Ask:** query the selected data. **Model & data:** inspect holdout metrics, calibration, quality exceptions and monitoring.

Embedded AI and the human step

LLM Q&A: submitting a question calls OpenAI for a restricted query plan. Python validates the plan, calculates all figures and displays the formula, source and filters. Only the question, schema/category catalogue and filters leave the server. **False positive:** wrong interpretation of a valid metric. **False negative:** unnecessary decline. The analyst checks interpretation and provenance. Missing API key disables Q&A visibly.

Anomaly detection: Isolation Forest flags the most unusual 3% of peer patterns; rules also flag DPD deterioration or bureau over-limit balances. 588 accounts trigger at least one rule/model flag. **False positive:** legitimate seasonality or stale bureau data. **False negative:** hidden deterioration resembling peers. Analysts confirm, dismiss or investigate with an evidence note; SQLite retains each decision. No anomaly labels are available to estimate accuracy.

Modelling and held-out results

Model	AUC	KS	Gini	Brier
Logistic regression	0.6857	0.2765	0.3714	0.1246
Gradient boosting	0.6603	0.2292	0.3206	0.1275

6,000/2,000/2,000 customers in train/validation/test; test has 2,948 loans. Bureau and repayment aggregates feed both models; imputation/encoding use training data only. Validation AUC selects gradient boosting (boosting 0.6829; logistic 0.6792). Logistic regression is stronger on test; selection remains frozen. Permutation importance uses validation data. Scores are retrospective; collections, IDs and protected attributes are excluded.

Three portfolio insights and actions

Below $30k: 38.67% (174/450) defaulted; $100k+: 9.52% (295/3,098). Prioritize affordability checks and early support for lower-income borrowers; income alone is not an approval rule.

Mortgage represents 52.31% of original exposure ($314,473,751) and has a 24.65% default rate. Review concentration limits and collateral valuations. Default counts alone miss the dollar exposure.

Unemployed, income <30k: 44.30% (101/228); highest among cross-segments with at least 100 accounts. Test targeted outreach in this segment and monitor results. This is an exploratory association, not a policy cutoff.

Assumptions and production needs

No default date, observation timestamp or prediction horizon is supplied. Bureau records may postdate loans; 3,875 collections events predate origination and are excluded from valid recovery totals. Original exposure is not loss/EAD. IT/data teams must provide dated snapshots and matured outcomes, agreed feature windows, secure secrets, durable review storage, access controls and independent validation. Monitor daily data quality, weekly score/feature drift and analyst outcomes, then AUC/KS/Gini and calibration on matured cohorts. API live-key, public hosting and video remain handover steps.